

ROUSHAN KUMAR

AI Training & Systems Engineer | Code Evaluation & Reasoning Specialist

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PROFESSIONAL SUMMARY

AI Training & Systems Engineer specializing in code quality evaluation, automated sandbox execution, and synthetic dataset generation. Experienced in parsing multi-language codebases using Tree-Sitter compilers, implementing OS runtime isolation for safe code execution, and authoring evaluation benchmarks for AI model reasoning. Skilled in building automated Python ETL pipelines, normalizing structural code features, and validating complex AI model outputs.

TECHNICAL SKILLS

AI Evaluation & Training: Code Quality Benchmarking • Synthetic Dataset Generation • Prompt Evaluation • Reinforcement Alignment

Parsing & Sandboxing: Tree-Sitter AST Compilers • Subprocess Sandbox Isolation • Syntax Analysis • ChromaDB Indexing

Backend & ETL: Python (FastAPI) • Node.js • Express.js • SQLite ETL Pipelines • Pandas • NumPy

Languages: C • Rust • Python • TypeScript • JavaScript • C# • Java • SQL

Testing & DevOps: PyTest • Jest • xUnit • Docker • CI/CD • GCP • Git • Linux Runtimes

Core Concepts: AST Graph Parsing • Code Execution Interception • Zero-Knowledge Crypto • Data Normalization

TECHNICAL PROJECTS

Astra Vision - Developer Sandbox & Code Graph Parser (FastAPI, Python, Monaco, Tree-Sitter)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

IDBI FinSync - AI-Powered Wealth & Financial Management Engine (Next.js, React, Fastify, Gemini API)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Co-created an intelligent personal financial management web app in a monorepo that unifies bank ledgers, investment portfolios, and expense streams into a live interactive dashboard.
- AI Integration Challenge:** Embedded "Mitra," an interactive AI wealth consultant leveraging Gemini API to analyze spending habits, detect budget anomalies, and deliver personalized financial guidance.
- Database & Concurency:** Designed PostgreSQL transaction ledgers validated by Zod schemas and Fastify microservice endpoints, handling concurrent balance updates with zero data loss.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- Cryptographic Engineering:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Engineering Virtual Simulation

June 2026

- Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- Action Taken:** Extended C# Web API controllers with `$set` atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- Engineering Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Engineering Virtual Simulation

June 2026

- Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- Engineering Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Engineering Virtual Simulation

May 2026

- Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (`<<`, `>>>`) and built 3NF database schemas with design pattern abstractions.
- Engineering Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using `csv` and `sqlite3` to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida

Expected July 2028

- Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)